

VISHAL KHARBANDA

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SUMMARY

Data & Backend Engineer with **3.5+ years** of experience in building **ETL/data pipelines**, Python-based microservices, and real-time data processing systems. Skilled in Spark, Kafka, Debezium, Azure data services, and SQL/NoSQL databases. Experienced in transforming and parsing large-scale raw data using Python and regex, with a strong foundation in CI/CD, Docker, and production deployments.

TECHNICAL SKILLS

Data Engineering: ETL Pipelines, Apache Spark, PySpark, Apache Kafka, Debezium, Apache Airflow, Data Warehousing, Data Modeling, Regex, Data Transformation

Databases: MongoDB, ArangoDB, RethinkDB, SQL Server, PostgreSQL, Azure Synapse, SQL Pool

Languages & Frameworks: Python, SQL, Flask, FastAPI, Node.js, Express, Java, REST APIs, OOP, Unit Testing

Cloud & DevOps: AWS, Azure, Docker, GitHub Actions (CI/CD), NGINX, PM2, SSH, Agile/Scrum

Tools: Git, Power BI, VS Code, Postman, Remote Monitoring, Structured Logging

PROFESSIONAL EXPERIENCE

Tech Mahindra, Noida — Software Engineer

Dec 2023 – Present

- Built **Python data transformation pipelines** using regex to parse and structure thousands of raw device packets from Qualcomm chipsets, supplying clean, structured data to the automation testing team.
- Optimized data parsing and transformation logic, **improving data pipeline efficiency by 40%** and reducing processing time for large-scale packet datasets.
- Designed and maintained scalable **Python-based microservices** for a mobile device testing platform, supporting distributed data collection across Linux, Windows, and iOS environments.
- Created and optimized **RESTful APIs** in Python (Flask/FastAPI) and Node.js (Express), enabling efficient data flow and real-time device telemetry.
- Implemented **CI/CD pipelines** using GitHub Actions for automated builds, testing, and deployments across NGINX-based Zoho servers.
- Operated, monitored, and troubleshoot distributed backend services across **80+ production servers**, using PM2, SSH, logging, and monitoring tools to ensure high availability.

XenonStack, Mohali, India — Data Engineer

Aug 2022 – Sep 2023

- Designed and implemented **real-time ETL pipelines** using Debezium and Azure services, reducing data latency and improving system availability.
 - Built **Kafka-based data ingestion pipelines** for collecting, streaming, and storing high-throughput wind turbine sensor data, enabling reliable real-time data flow for downstream analytics.
 - Processed and analyzed large-scale datasets using **Apache Spark and PySpark**, enabling efficient distributed data processing and transformation for analytics workloads.
 - Developed a **UPI fraud detection system** leveraging ArangoDB for secure and efficient pattern-based anomaly detection on transaction data.
 - Built and integrated **Power BI dashboards** with Azure Synapse and SQL Pool, **boosting analytics performance by 20%** and enabling faster, data-driven insights.
 - Collaborated with cross-functional teams to streamline data ingestion and quality processes, **reducing integration time by 20%** and enhancing overall data reliability.
 - Gained working knowledge of **Apache Airflow** for workflow orchestration and scheduling of batch data pipelines.
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PROJECTS

UPI Fraud Detection System — XenonStack, Mohali

Jan 2023 – Sep 2023

- Developed a **real-time fraud detection system** for UPI transactions using ArangoDB and pattern-based anomaly detection techniques.

- Built **event-driven ETL pipelines** using Debezium and Azure services to enable near real-time transaction monitoring and alerting.
- Architected optimized **data models** to identify suspicious behavioral patterns, improving fraud detection accuracy and response time.

TMDC (Tech Mahindra Device Cloud) — Tech Mahindra, Noida

Jan 2024 – Present

- Built **Python data pipelines** for device telemetry collection, transforming raw packet data into structured formats consumed by automation and analytics teams.
 - Designed **RESTful APIs** for secure, low-latency data exchange between cloud services and physical devices; improved observability via structured logging and monitoring.
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CERTIFICATIONS

- **Generative AI for Beginners** — Udemy (Aakriti E-Learning Academy), May 2026
Covered foundational concepts of Generative AI including prompt engineering, LLMs, and AI-driven automation.
 - **AI White Belt** — Tech Mahindra (SPOT Certificate)
Recognized for demonstrated AI technology proficiency and commitment to AI-first strategy; awarded for applied AI skills in enterprise workflows.
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EDUCATION

Bachelor of Technology (B.Tech.) — Computer Science Engineering

I.K. Gujral Punjab Technical University, Jalandhar | Chandigarh Group of Colleges, Landran

Graduated May 2022 | GPA: 8.2